Abhi Muvva

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Education

Sep. 2024 – May 2026

University of Wisconsin–Madison – M.S. in Physics – Quantum Computing — Madison, WI

- **Relevant Coursework:** Quantum Algorithms, Quantum System Architecture, Programming Languages for Quantum Computers, Quantum Computing Lab, Qubit Tune Up and Programming, Advanced Quantum Computing, Intro to Quantum Computing and Quantum Mechanics.
- **Cumulative GPA:** 3.73 / 4.00

Aug. 2020 – May 2024

Mahindra University – B.Tech. in Artificial Intelligence — Hyderabad, India

- **Relevant Coursework:** Quantum Computing, Machine Learning, Neural Networks and Deep Learning, Natural Language Processing, Digital Image Processing.
- **Honors:** Founded the Quantum Computing Club (Qubit); Vice Head of the Astronomy Club (Celeste).

Experience

Dec. 2025 – Present

Graduate Researcher – Dr. Matthew Otten, University of Wisconsin–Madison — Madison, WI

- Replicated **QFlow(4e,4o)** from primary literature (Kowalski & Bauman, 2023), achieving <0.1 mHa agreement with published H_6/H_8 benchmarks across equilibrium and stretched geometries.
- Developing a **fragmentation framework** that decomposes a 36-orbital **Fe_4S_4** active space into overlapping orbital subsets solved by **TrimCI** (a selected-CI solver), with self-consistent variational coupling between fragments, benchmarking whether QFlow-style fragmentation reduces the classical computational cost for strongly correlated electronic structure.
- Implemented **DUCC downfolding** infrastructure (GCCSD spin-orbital amplitude extraction, anti-Hermitization, σ_{ext} partitioning) for coupled-cluster dressed effective Hamiltonians in reduced target spaces, following Bauman et al. (JCP 151, 2019).

Jun. 2023 – Jul. 2023

Research Intern, QMatter Labs — Hyderabad, India

- Prototyped a **grid-based discretization** for EV charging station placement where each grid cell maps to one qubit, letting the problem scale directly with available quantum hardware.
- Built a proof-of-concept hybrid solver pairing a classical **Genetic Algorithm** with a basic Qiskit quantum circuit. The grid encoding worked as expected, but the quantum component produced unreliable results, which drove the deeper algorithm redesign in later work.

Jul. 2023 – Oct. 2023

Visiting Student Researcher – CyberVSR, Rochester Institute of Technology — Rochester, NY

- Developed a deep-learning password guessing model using **WGAN-GP**; collected, cleaned, and visualized leaked-password datasets for training and evaluation.

Projects

Ongoing

Classical QFlow with TrimCI Sub-Solver for Fe_4S_4 — *Python, PySCF, TrimCI, NumPy/SciPy*

- Identified that QFlow's effective Hamiltonian construction is intractable for the **Fe_4S_4** system ($\sim 7 \times 10^{13}$ determinants), motivating a redesign from wavefunction-space dressing to **integral-level embedding**.
- Designing an orbital fragmentation engine using sliding-window decomposition over energy-ordered orbitals, with tunable fragment size (12–18 orbitals) and overlap, targeting the regime where individual fragments are tractable for **TrimCI** while the full 36-orbital space requires $\sim$10,000 determinants.
- Building a self-consistent coupling layer where each fragment's one-particle reduced density matrix (**1-RDM**) dresses neighboring fragments' one-electron integrals via Coulomb and exchange corrections, iterated to convergence.
- Evaluating total determinant count across fragment solves against brute-force TrimCI on the full active space as a direct measure of fragmentation benefit for a system at the edge of classical tractability.

Mar. 2026

QFlow Algorithm Replication and Validation — *Python, PySCF, PennyLane, NumPy/SciPy*

- Built a modular **QFlow(4e,4o)** implementation from the primary literature (arXiv:2305.05168v2), including **SES enumeration**, **CAS basis construction** (36-determinant, 35-parameter blocks), **effective Hamiltonian dressing** via sparse matrix exponentials, and **commutator-gradient** optimization with first-claimant amplitude ownership.

- Validated on four benchmark cases (H_6 and H_8 at $R = 2.0$ and $R = 3.0$ a.u. in STO-3G), converging to <0.1 mHa of published energies and recovering $>97\%$ of correlation energy relative to full-space exact diagonalization.
- Diagnosed and resolved multiple implementation pitfalls not addressed in the paper: prefactor errors in two-electron integral absorption ($\texttt{fac} = 0.5$ vs 1.0), spin-summation double-counting in H_{eff} construction, and convergence sensitivity to the interplay between energy spread and gradient thresholds.

Feb. 2026 – Present

HGQA: Hybrid Quantum-Classical EVCS Optimizer — *Python, Qiskit, Qiskit Aer*

- Reformulated EV charger placement as a **six-term QUBO** (H_1–H_6) with Chebyshev-distance pairwise terms and radius-cutoff sparsity. Designed a **three-tier cell pruner** (dead-cell removal, greedy lower-bound elimination, spatial deduplication) that cuts qubit count by **50–99%**.
- Configured **MPS-based QAOA simulation** through Qiskit Aer for 80–100+ post-prune qubits in tensor-network form; identified and corrected a large identity offset ($\sim$1500) that suppressed the actual cost signal ($\sim$8), restoring meaningful COBYLA convergence.
- Built the end-to-end **data-to-QAOA pipeline** with per-term QUBO normalization, post-build penalty calibration, and sparsity-preserving Ising conversion. Implemented multiple population generators (random, greedy-with-perturbation, QAOA-seeded) under a common interface for controlled ablation studies.

Dec. 2025

Zero-Noise Extrapolation with Randomized Benchmarking — *Python, QUA, Quantum Machines*

- Designed a unified RB+ZNE protocol for five superconducting qubits as part of a graduate course collaboration with Qolab (co-founded by **John M. Martinis**), extracting intrinsic error per Clifford and gate-specific noise sensitivity via controlled noise amplification across $\lambda \in \{1, 3, 5\}$.
- Extended Qolab's **QUA**-based RB framework with gate-family-specific noise scaling: **digital Clifford folding** for microwave X/Y gates and **analog pulse stretching** for flux-driven Z rotations, addressing the broken $G^{\dagger}G$ symmetry in nonlinear flux-bias trajectories.
- Extracted zero-noise gate fidelities of $f_0^{(X)} \approx 0.9997$ (microwave) and $f_0^{(Z)} \approx 0.9983$ (flux) via linear extrapolation of EPG(λ), establishing a quantitative fidelity hierarchy between the two gate families.

Nov. 2025

Neutral-Atom Quantum Processor Simulator — *Python, QuTiP, Qiskit*

- Built a **pulse-level** simulator of a neutral-atom processor using a four-level atomic model evolved under the **Lindblad master equation**, capturing **Rydberg excitation**, spontaneous decay, leakage to non-computational states, and **blockade dynamics** between neighboring atoms.
- Implemented calibrated RX, RZ, and **Rydberg-mediated CZ gates** using explicit time-dependent control pulses, with circuit execution integrated into a Qiskit-compatible workflow for algorithm benchmarking.
- Validated gate fidelities via single- and two-qubit **randomized benchmarking** and **interleaved RB**, reproducing the expected fidelity hierarchy between high-fidelity single-qubit gates and noisier CZ operations.

Oct. 2025

Magneto-Optical Trap (MOT) – Rb-87 — *Experimental Quantum Optics Practicum*

- Reconstructed and re-aligned a full **Rb-87 MOT** setup from a misaligned state, optimizing beam paths, polarization purity, and retro-reflections for stable atom trapping.
- Calibrated and frequency-locked a diode laser via **saturated absorption spectroscopy**, analyzing hyperfine transitions and error signals for stable detuning control.

Apr. 2025

WISQ Compiler Enhancement — *Python, Qiskit, Q#*

- Built a **Hellinger fidelity** verification pass and noisy error-rate comparison for the WISQ compiler, confirming functional equivalence (fidelity = 1.0) between original and GUOQ-optimized circuits and quantifying a **5.8% error-rate reduction** under the FakeParis noise model.
- Integrated **Q#-Estimator** resource profiling into the compilation pipeline, measuring a **30% T-state reduction** (647→450), **25% logical-depth reduction** (455→342), and **6% physical-qubit savings** across benchmark circuits, with an automated harness showing gains scale with longer GUOQ optimization timeouts.

Technical Skills

Languages:	Python, MATLAB, C++, MySQL
Quantum Frameworks:	QuTiP, Qiskit, PySCF, PennyLane, Cirq, Qualtran, openFermion, QUA, Qrisp, OpenQASM3, Q#
Scientific Computing:	NumPy, SciPy, Pandas, Matplotlib, TensorFlow
Development & Tools:	Git, Docker, Linux/Bash, Jupyter, VS Code, Anaconda, HPC job submission

Activities

Hackathons: MIT-iQuISE, iQuHACK-2024; Catalog, Full Day Hackathon-2023; Hacktoberfest-2023, 2022.

Workshops: Conducted multiple Quantum Computing and QML workshops in collaboration with QWS; Conducted a full-day Introduction to Quantum Computing workshop at Gitam University.

References

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